

Maoqi Liu

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Education

Beijing University of Posts and Telecommunications	2025.09 – Present
• M.S. in Intelligent Science and Technology	Advisor: Quan Fang
Beijing University of Posts and Telecommunications	2021.09 – 2025.06
• B.Sc. in Artificial Intelligence	

Publications

1. NOTAM-Evolve: A Knowledge-Guided Self-Evolving Optimization Framework with LLMs for NOTAM Interpretation

First Author Accept by AAAI2026 Conference (CCF-A)

- Overview:** Accurate interpretation of Notices to Airmen (NOTAMs) is vital for aviation safety, yet their condensed and cryptic language challenges both humans and automated systems. We define the task as deep parsing, which requires dynamic knowledge grounding and schema-based inference. To address this, we propose NOTAM-Evolve, a knowledge-guided self-evolving framework that enables large language models to autonomously enhance their reasoning through a closed-loop learning process with knowledge graph – based retrieval. Experiments on 10,000 expert-annotated NOTAMs show a 30.4% accuracy gain over base LLMs, establishing a new state of the art in structured NOTAM interpretation.

2. Knots: A Large-Scale Multi-Agent Enhanced Expert-Annotated Dataset and LLM Prompt Optimization for NOTAM

Semantic Parsing

First Author Accept by Advanced Engineering Informatics Journal (CCF-B, SCI-Q1)

- Overview:** Notices to Air Missions (NOTAMs) convey essential flight safety information but remain difficult to parse due to their complex structure and implicit semantics. We introduce NOTAM semantic parsing, a task focused on deep understanding through semantic inference and aviation knowledge integration. To support it, we build Knots, a dataset of 12,347 expert-annotated NOTAMs across 194 Flight Information Regions, developed via a multi-agent framework for comprehensive field discovery. Extensive experiments on prompt design and model adaptation show notable gains in aviation text comprehension, offering new insights for automated NOTAM analysis.

3. R²A²-MoE: Ridge Regression-based Analytic Adaptation with Mixture of Experts for Continual Learning with Vision-

Language Models

First Author Accept by PRCV2025 Conference (CCF-C, Oral)

- Overview:** Vision-Language Models (VLMs) show strong zero-shot performance but struggle to retain it under continual learning due to dual forgetting. We propose R²A²-MoE—Ridge Regression-based Analytic Adaptation with Mixture of Experts—which enables efficient collaborative specialization through closed-form expert optimization. This approach mitigates both backward and forward forgetting while improving adaptability. Experiments on X-TAIL and MTIL benchmarks demonstrate superior performance and efficiency, reducing parameters, memory, and training time by over 45%, 56%, and 46%, respectively.

Projects

Job Recommendation System Based on Knowledge Graphs Project Leader	2024.04 – 2024.06
• Overview: This project leverages a knowledge graph to help university students quickly identify ideal job opportunities and clarify their career positioning. By fine-tuning the open-source Paddle large language model for skill extraction and quantitative evaluation, the project fills gaps in job-related knowledge graphs and enhances visual analytics for understanding employment trends.	

Awards

Beijing Natural Science Foundation	Undergraduate “QiYan” Research Grant	2024
2022 National English Competition for College Students	Second Prize	2022.10

Personal Statement

- Demonstrated leadership and teamwork as Vice President of BUPT’s Long-distance Running Association, debate team champion, and class leader at both undergraduate and graduate levels.
- Over 100 hours of volunteer service, actively supporting community initiatives and providing daily assistance to fellow students.